VB Environment

Visual Basic 6.0 IDE
OAVS Computer Teacher - Concise Study Notes

Syllabus topic (a. VB Environment)	Covered in section
i. Menu Bar, Toolbars, Tool Box	1. Menu Bar | 2. Toolbars | 3. Tool Box
ii. Project Explorer	4. Project Explorer
iii. Properties Window	5. Properties Window
iv. Form Designer	6. Form Designer
v. Form Layout (printed "From Layout" in the syllabus)	7. Form Layout

IDE at a glance

The **VB 6.0 IDE (Integrated Development Environment)** is the single workspace in which a Visual Basic application is designed, built, run and saved. Everything in this document is a part of that workspace. The seven numbered parts below are the interface elements you must be able to identify, label and describe.

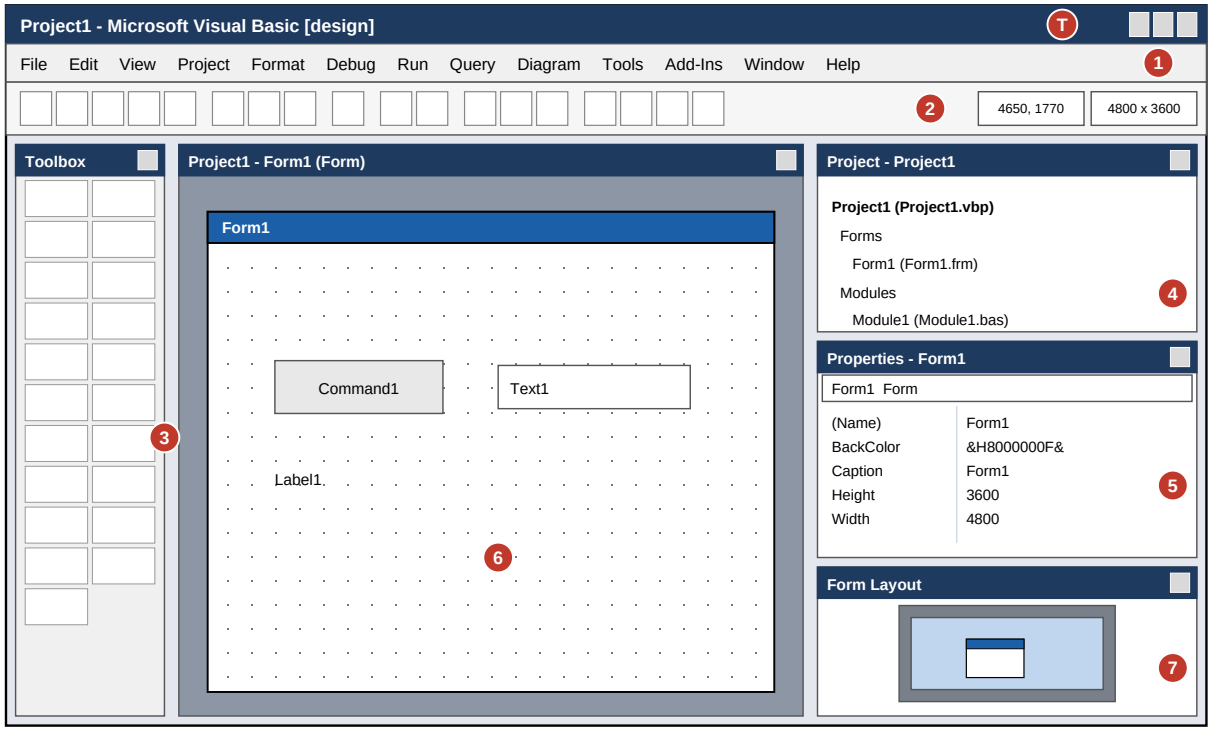


Fig. 1: VB 6.0 IDE (Environment) overview - T = Title Bar; numbered components are described in Sections 1-7

No.	Component	Position in default layout	One-line role
T	Title Bar	Very top of the IDE window	Shows project name and current mode [design] / [run] / [break]; has Minimize, Maximize, Close buttons
1	Menu Bar	Below the Title Bar (top)	Drop-down menus giving access to all IDE commands
2	Toolbars	Below the Menu Bar	Buttons that give one-click shortcuts to common commands
3	Tool Box	Left edge	Palette of tools/controls to draw on the form
4	Project Explorer	Top right	Tree view of all files/components in the project
5	Properties Window	Middle right	Lists and edits properties of the selected object
6	Form Designer	Centre	Visual window where the form (user interface) is designed
7	Form Layout	Bottom right	Miniature screen to set form start-up position

OAVS Exam Points - IDE

- The **Title Bar** shows the current mode: **[design]** (building), **[run]** (application running), **[break]** (execution paused).
- Default new-project type in the New Project dialog is **Standard EXE**.
- VB 6.0 IDE can be set to **MDI** (default - windows float inside one parent window) or **SDI** (Tools > Options > Advanced tab).

1 Menu Bar

Definition and Purpose

The **Menu Bar** is the horizontal bar of drop-down menus placed directly below the Title Bar. It is the main command centre of the IDE: nearly every action - creating, opening, saving, designing, running and managing a project - can be reached from it.

Main Components - the 13 Menus of VB 6.0

Menu	Purpose / typical commands
File	New / Open / Add / Remove / Save project, Save Form, Print, **Make .exe**, Exit
Edit	Undo, Redo, Cut, Copy, Paste, Delete, Select All, Find / Replace
View	Show or hide IDE windows: Code, Object, Object Browser, Project Explorer, Properties Window, Form Layout Window, Toolbox, Toolbars
Project	Add Form / Module / Class Module / User Control, **Components**, References, Project Properties
Format	Arrange controls on a form: Align, Make Same Size, Spacing, Center in Form, Order, Lock Controls
Debug	Tools for testing/tracing a running application
Run	**Start (F5)**, Start With Full Compile, Break, End, Restart
Query	Data-tool menu for query design (shown with the data tools)
Diagram	Data-tool menu for database diagrams (shown with the data tools)
Tools	Add Procedure, **Menu Editor (Ctrl+E)**, **Options** (environment settings)
Add-Ins	**Add-In Manager** - load / unload add-ins
Window	Split, Tile Horizontally / Vertically, Cascade, Arrange Icons, list of open windows
Help	MSDN Library, Contents, Index, Search, About Microsoft Visual Basic

Important Interface Elements

- **Access keys** - underlined letter, used with Alt (e.g. Alt+F opens File).
- **Shortcut keys** shown at the right of a command (e.g. F5, Ctrl+N).
- **Ellipsis (...)** = dialog box follows; **arrow** = sub-menu; **greyed** item = unavailable now.
- Right-clicking a window opens a context (pop-up) menu with commands for that item.

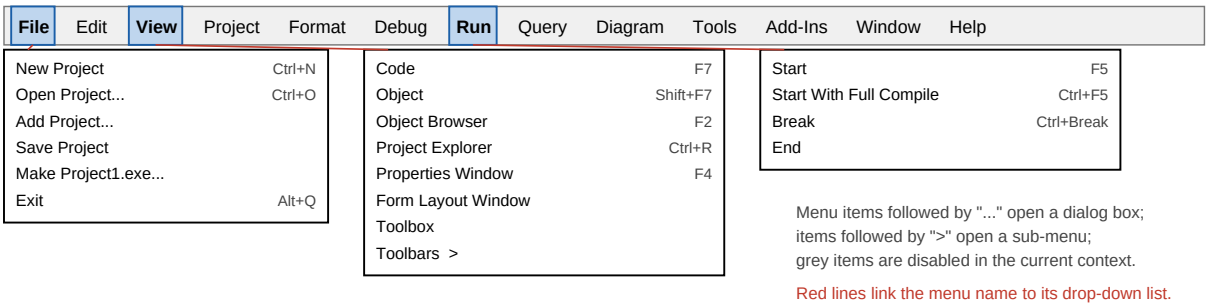


Fig. 2: VB 6.0 Menu Bar with sample File, View and Run drop-down menus

Functions and Uses in VB 6.0

- Create (**File > New Project**), open and save projects; build the executable (**File > Make .exe**).
- Open or hide IDE windows (**View** menu) - e.g. Properties Window (F4), Project Explorer (Ctrl+R).
- Add forms/modules (**Project > Add Form**) and extra controls (**Project > Components**, Ctrl+T).
- Run the application with **Run > Start** (F5); stop with **Run > End**.
- Align and size controls on the form with the **Format** menu (active only while Form Designer is active).

OAVS Exam Points

- Menu Bar has **13 menus**: File, Edit, View, Project, Format, Debug, Run, Query, Diagram, Tools, Add-Ins, Window, Help.
- **F5** = Start, **F4** = Properties Window, **Ctrl+R** = Project Explorer, **Ctrl+T** = Components, **Ctrl+E** = Menu Editor, **F7** = Code window, **F2** = Object Browser.
- Environment settings (e.g. grid) are changed from **Tools > Options**, not from the Project menu.

Summary

Menu Bar = row of 13 drop-down menus below the Title Bar; the full command set of the IDE.

2 Toolbars

Definition and Purpose

Toolbars are rows of small icon buttons placed below the Menu Bar. Each button is a one-click shortcut to a command that is also available in a menu, so common work is done faster. Pointing at a button shows its name as a **ToolTip**.

Four built-in toolbars (View > Toolbars, or right-click on any toolbar):

Standard	Edit	Debug	Form Editor
Project, edit, run and IDE-window buttons	Code-editing helpers (for Code window)	Run/break/step tools for testing	Align, size, order, lock controls

Fig. 3: The Standard Toolbar (button groups) and the four built-in toolbars

Main Components - Built-in Toolbars

Toolbar	Purpose	Shown by default?
Standard	Project, clipboard, run and IDE-window buttons (general work)	Yes
Edit	Extra buttons for editing in the Code window	No
Debug	Buttons for running, breaking and tracing an application	No
Form Editor	Bring to Front, Send to Back, Align, Center, Make Same Size, Lock Controls	No

Important Interface Elements - Standard Toolbar Buttons

Button group	Buttons and function
File / Project	Add Project (add new project to group), Add Form, **Menu Editor**, Open Project, Save Project
Clipboard / Find / Undo	Cut, Copy, Paste, Find; Undo, Redo
Run	**Start** (green triangle) runs the app; **Break** (two bars) pauses; **End** (red square) stops
IDE Windows	Project Explorer, Properties Window, Form Layout Window, Object Browser, Toolbox, Data View Window, Visual Component Manager
Position & Size	Right end of toolbar: Left/Top position and Width x Height of the selected form or control (in **twips**)

Functions and Uses in VB 6.0

- Open or close toolbars: **View > Toolbars** or right-click any toolbar; **Customize** adds/removes buttons.
- Toolbars can be **docked** to an edge or dragged to float as a separate palette.
- The Windows buttons at the right of the Standard toolbar re-open Project Explorer, Properties Window, Form Layout and Toolbox if they were closed.

OAVS Exam Points

- VB 6.0 has **four** built-in toolbars: Standard, Edit, Debug, Form Editor.
- Only the **Standard** toolbar appears by default.
- Run buttons on toolbar = **Start, Break, End** (same as Run menu).
- Position/size boxes show values in **twips** (1 inch = 1440 twips).

Commonly Confused

- **Standard** toolbar (project/run buttons) vs **Form Editor** toolbar (layout buttons for controls).
- **Toolbar** (command buttons) vs **Toolbox** (controls to draw) - different windows.

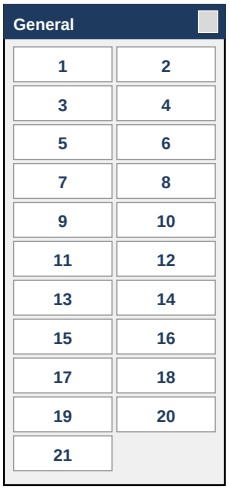
Summary

Toolbars = icon-button shortcuts to menu commands; 4 built-in, Standard shown by default.

3 Tool Box

Definition and Purpose

The **Toolbox** is a floating/dockable window (default: left side of the IDE) that holds the **tools** used to place objects on a form. You choose a tool and draw it on the form in the Form Designer; each drawn item is a **control**.



Tool names (number = position in the Toolbox, left to right, top to bottom)

1. Pointer
2. PictureBox
3. Label
4. TextBox
5. Frame
6. CommandButton
7. CheckBox
8. OptionButton
9. ComboBox
10. ListBox
11. HScrollBar
12. VScrollBar
13. Timer
14. DriveListBox
15. DirListBox
16. FileListBox
17. Shape
18. Line
19. Image
20. Data
21. OLE

21 icons = 1 Pointer + 20 standard (intrinsic) controls.
Pointer only selects/moves/resizes objects - it does not draw anything.

Fig. 4: Toolbox (General tab) with all 21 standard tools

Main Components and Features

Component	Description
Pointer	Only selection tool (not a control). Selects, moves and resizes objects.
Intrinsic (standard) controls	20 controls built into VB; always present in a Standard EXE project.
ActiveX controls	Extra controls added through **Project > Components (Ctrl+T)** or right-click > Components.
General tab	Default tab that holds the standard tools; extra **tabs** can be added by right-click > Add Tab.
Title bar / Close button	Toolbox can be docked, floated or closed; reopen via **View > Toolbox** or Standard toolbar button.

All Toolbox Tools with Their Functions (Grouped)

Group	Tool	Function
Text / display	Label	Shows text that the user cannot edit
	TextBox	Accepts or displays text typed by the user
	PictureBox	Displays pictures; can also hold (contain) other controls
	Image	Displays a picture (lighter than PictureBox; cannot contain controls)
Selection	CheckBox	On/Off choice; several can be ticked together
	OptionButton	Choose only one option from a group
	ComboBox	Drop-down list combined with a text box
	ListBox	Displays a list of items to choose from
Action / container	CommandButton	Button clicked by the user to perform an action
	Frame	Container that groups related controls (e.g. option buttons)
Scrolling	HScrollBar	Horizontal scroll bar to move through a range of values
	VScrollBar	Vertical scroll bar to move through a range of values
Disk / file	DriveListBox	Displays and selects disk drives
	DirListBox	Displays the directories (folders) of the selected drive
	FileListBox	Lists the files of the selected folder
Drawing	Shape	Draws a rectangle, square, oval, circle or rounded shape
	Line	Draws a straight line
Other	Pointer	Selection tool only (not a control): selects, moves and resizes objects
	Timer	Triggers an action at set time intervals; invisible at run time
	Data	Connects a form to a database file; provides record-navigation buttons
	OLE	Embeds or links objects from other applications (e.g. a worksheet)

Functions and Uses in VB 6.0

- **Double-click** a tool: a default-size control appears in the centre of the form.
- **Click the tool, then drag** on the form: draws the control at the chosen size and place.
- **Ctrl + click** a tool to draw several controls of the same type without re-selecting it.
- The Toolbox shows tools only when a **form (Form Designer)** is active.
- Each new control gets a default name such as Command1, Text1, Label1.

OAVS Exam Points

- Toolbox shows **21 icons** = Pointer + **20** standard controls.
- The **Pointer** is not a control.
- To add more controls (ActiveX): **Project > Components** - shortcut **Ctrl+T**.
- **Timer** is placed on the form at design time but is not visible at run time.

Commonly Confused

- **Toolbox** (what you draw with) vs **Toolbar** (command shortcuts).
- **Intrinsic** controls (built-in) vs **ActiveX** controls (added by Components dialog).

Summary

Toolbox = palette of 21 tools (Pointer + 20 controls); ActiveX controls are added via Ctrl+T.

4 Project Explorer

Definition and Purpose

The **Project Explorer** window displays a **hierarchical (tree) list of the project and every component it contains** - forms, modules, class modules and so on. A group of several projects (a **project group**) is shown together. It is used to see, open and manage the files that make up an application.

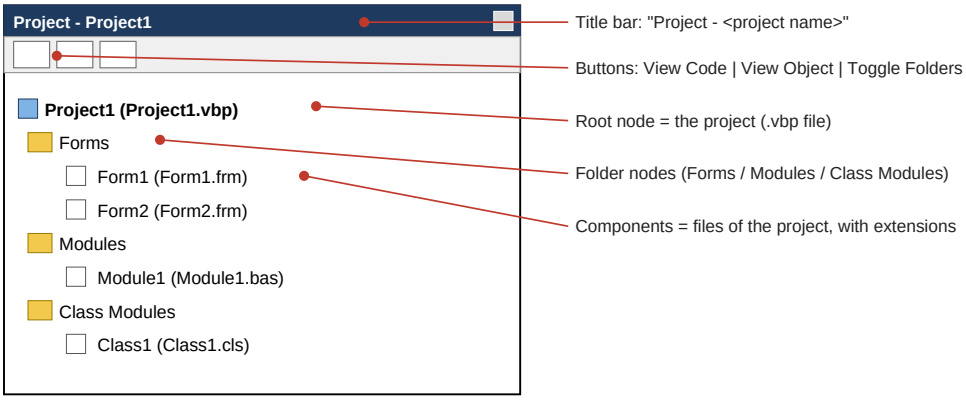


Fig. 5: Project Explorer window with labeled parts

Main Components and Interface Elements

Element	Function
View Code button	Opens the Code window of the selected item
View Object button	Shows the selected object (e.g. the form in Form Designer)
Toggle Folders button	Switches between folder view and plain list of components
Project node	Root of the tree - the project (or project group) name
Folders	Forms, Modules, Class Modules (and others) group similar components
Component names	Shown as *Name (FileName.ext)*

Project Files and Extensions (high-yield)

Extension	File type	Extension	File type
.vbp	Visual Basic **Project**	.bas	Standard **Module**
.vbg	Project **Group**	.cls	**Class** Module
.frm	**Form** file	.ctl	User **Control**
.frx	Binary data of a form (pictures, icons)	.res	Resource file

Functions and Uses in VB 6.0

- Open the Project Explorer: **View > Project Explorer**, **Ctrl+R**, or the Standard toolbar button.
- **Double-click** a form to open it in the Form Designer; select an item and use View Code / View Object.
- Right-click an item to **Add**, **Remove**, **Save** or set properties of the project/components.
- Shows exactly which files belong to the application, so all must be kept together when saving or copying.

OAVS Exam Points

- Project Explorer = **tree view of project components**; default position: top right.
- Project file = **.vbp**; Form = **.frm**; Module = **.bas**; Class = **.cls**; Group = **.vbg**.
- It has **3 buttons**: View Code, View Object, Toggle Folders.

Commonly Confused

- **.frm** (form design file) vs **.frx** (binary data of the form).
- **.vbp** (single project) vs **.vbg** (group of projects).

Summary

Project Explorer = tree of the project's forms, modules and classes; opened with Ctrl+R.

5 Properties Window

Definition and Purpose

The **Properties Window** lists all **properties** (attributes such as name, caption, colour, size) of the currently selected object - the form or any control on it - and lets you view or change their **settings** at **design time**.

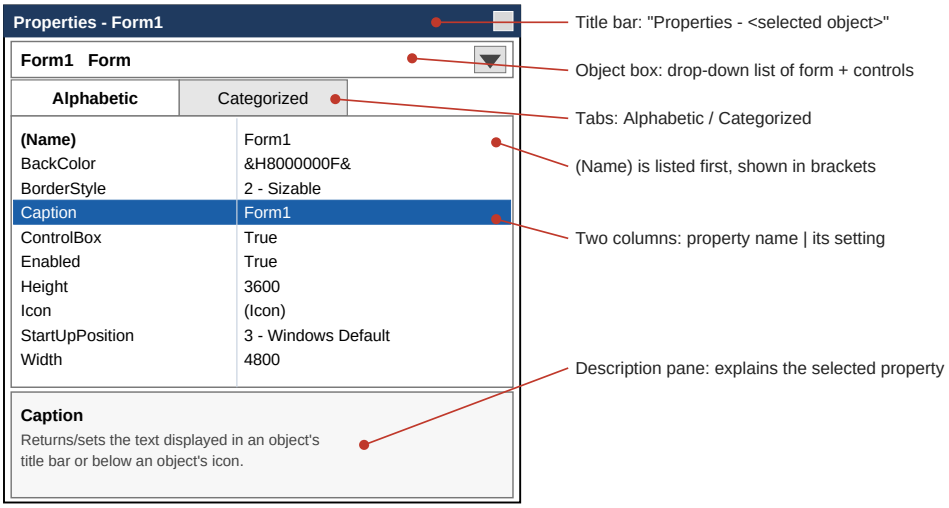


Fig. 6: Properties Window (Alphabetic tab) with labeled parts

Main Components and Interface Elements

Element	Function
Title bar	Displays "Properties - *ObjectName*"
Object box	Drop-down list of the form and all its controls; choose the object to work on
Alphabetic tab	Properties in A-Z order, **(Name)** first
Categorized tab	Properties grouped by category (Appearance, Behavior, Font, Position, etc.)
Properties list	Two columns: **property name** (left) and **setting** (right)
Description pane	Bottom area explaining the property currently selected

Frequently Seen Form Properties

Property	Meaning
(Name)	Name used to refer to the object (e.g. Form1, Command1). Shown in brackets.
Caption	Text shown in the title bar (form) or on the object (button, label).
BackColor	Background colour of the object
BorderStyle	Border type of the form (e.g. 2 - Sizable); decides if it can be resized
StartUpPosition	Where the form appears when the application starts
Icon / ControlBox	Icon in the title bar / whether the Control menu box is displayed
Height, Width, Left, Top	Size and position in twips

Functions and Uses in VB 6.0

- Open: **F4**, **View > Properties Window** or the Standard toolbar button.
- It always shows the properties of the **selected** object; select a different object and the list changes.
- Select a property, then **type** the value or pick it from a **drop-down list**; "..." button opens a dialog (e.g. colour, picture, font).
- If several controls are selected together, only their **common** properties are listed.

OAVS Exam Points

- Shortcut key: **F4**. Default position: middle right of IDE.
- Name rule: begins with a letter, up to **40 characters**, only letters, digits and underscore - no spaces.
- Properties set in this window take effect at **design time**.

Commonly Confused

- **(Name)** - identity used by VB; **Caption** - text shown to the user. They are different.
- Properties **Window** (the pane) vs **Properties** (the attributes listed inside it).
- **Project Explorer** (components of the project) vs **Properties Window** (attributes of one object).

Summary

Properties Window = list of attributes of the selected object, edited at design time; opened with F4.

6 Form Designer

Definition and Purpose

The **Form Designer** (also called the **Form window** or **Object window**) is the central window where the **visual user interface** of the application is created. A **form** is the window of the application; controls from the Toolbox are drawn on it and arranged exactly as the user will see them.

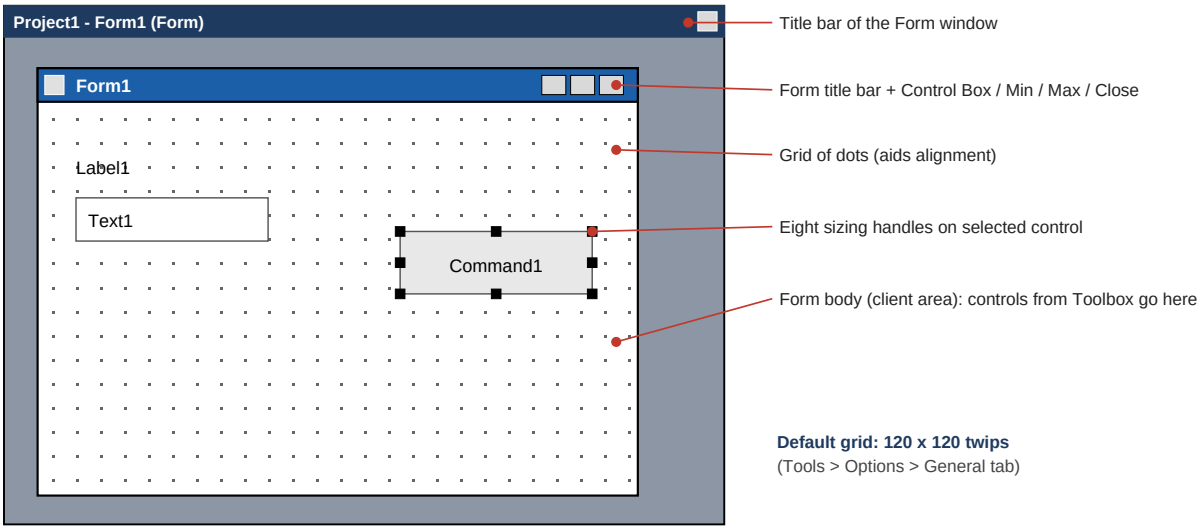


Fig. 7: Form Designer showing a form with grid dots, controls and sizing handles

Main Components and Interface Elements

Element	Function
Form window title bar	Shows "*ProjectName - FormName (Form)*"
Form (title bar, Control Box, Min / Max / Close)	Represents the application window; its Caption appears in the title bar
Grid (dots)	Dotted pattern that helps align and size controls (design time only - not shown at run time)
Sizing handles	Eight small squares around a selected control used to resize it
Form body (client area)	Container area on which controls are placed

Functions and Uses in VB 6.0

- Open: double-click the form in **Project Explorer**, or select it and click **View Object** (**View > Object**, Shift+F7).
- **Draw** controls from the Toolbox; **move** by dragging; **resize** with handles; **delete** with the Delete key.
- **Select several controls**: Ctrl + click or drag a selection rectangle; then use the **Format** menu (Align, Make Same Size, Center in Form, Lock Controls).
- Keyboard fine-tuning: **Ctrl + arrow** moves a control by one grid unit; **Shift + arrow** resizes it.
- Grid settings (Show Grid, Grid Units, Align Controls to Grid) are in **Tools > Options > General**.
- The Form Designer is the **design view**; the form is shown as the user sees it only after running (F5).

OAVS Exam Points

- Form Designer = place where **forms are visually designed**; the form is the container for controls.
- Default scale unit is the **twip** (1 inch = 1440 twips); default grid = **120 x 120 twips**.
- Selected control shows **8 sizing handles**.
- New project opens with one form named **Form1**.

Commonly Confused

- **Form Designer** (visual design of the form) vs **Code window** (separate window for writing code - not part of this topic).
- **Form Designer** (design the form) vs **Form Layout** (position the form on the screen).

Summary

Form Designer = visual design surface (form + grid) where controls are placed, moved and sized.

7 Form Layout (Form Layout Window)

Definition and Purpose

The **Form Layout Window** shows a **miniature computer screen** with a small picture of the form on it. By **dragging** this mini-form, you decide where the form will appear on the screen when the application starts - without running the program.

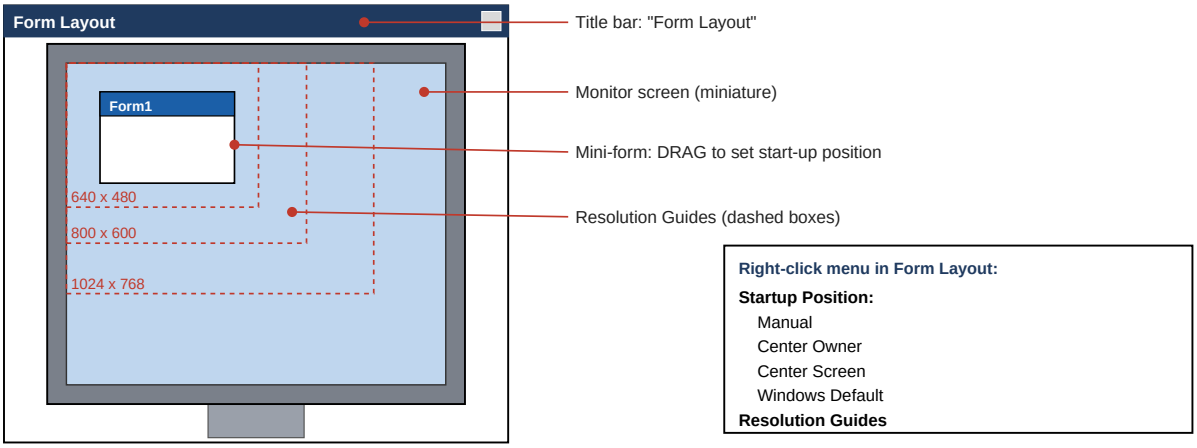


Fig. 8: Form Layout window with mini-form, resolution guides and right-click menu

Main Components and Interface Elements

Element	Function
Monitor screen	Miniature of the display on which the form is positioned
Mini-form	Small picture of the form; drag it to set the start-up position
Resolution Guides	Dashed rectangles showing screen sizes (e.g. 640 x 480, 800 x 600, 1024 x 768) to check how the form fits
Right-click menu	**Startup Position** options and **Resolution Guides** switch

Startup Position Options (right-click menu)

Option	Value	Effect
Manual	0	Form appears where it was placed in Form Layout (set automatically when you drag the mini-form)
Center Owner	1	Centred on the owner form (e.g. over its parent form)
Center Screen	2	Centred on the screen
Windows Default	3	Position chosen by Windows (default setting)

Functions and Uses in VB 6.0

- Open: **View > Form Layout Window** or the Standard toolbar button.
- Drag the mini-form to a position: the form's **StartUpPosition** becomes **Manual** and the position is saved with the form.
- Use **Resolution Guides** to design for users with smaller screens.
- Especially useful in projects with **many forms**, so each form opens at the right place.

OAVS Exam Points

- Form Layout Window = sets the form's **start-up position** on the screen.
- Its effect is stored in the form property **StartUpPosition** (shown in the Properties Window).
- Resolution Guides help design for different **screen resolutions**.

Commonly Confused

- **Form Layout** (where the form opens on screen) vs **Form Designer** (what is inside the form).
- Syllabus prints "From Layout" - the topic is **Form Layout**.

Summary

Form Layout = miniature screen where dragging the mini-form sets the form's start-up position.

QR Quick Revision

Key Definitions

Term	Definition (one line)
IDE	Integrated Development Environment - single workspace to design, build, run and save VB applications.
Menu Bar	13 drop-down menus below the Title Bar giving access to all IDE commands.
Toolbars	Rows of icon buttons that are shortcuts to menu commands (Standard, Edit, Debug, Form Editor).
Tool Box	Window of 21 tools (Pointer + 20 controls) used to draw controls on a form.
Project Explorer	Tree view of the project and its forms, modules and class modules.
Properties Window	Lists and edits properties of the selected object at design time.
Form Designer	Window for visually designing a form and placing controls on it.
Form Layout	Miniature screen window to set the form's start-up position.

Component Functions and Shortcuts

Component	Open with	Default location	Chief function
Menu Bar	Always visible	Under Title Bar	All commands
Toolbars	View > Toolbars	Under Menu Bar	One-click shortcuts
Tool Box	View > Toolbox	Left	Choose/draw controls
Project Explorer	Ctrl+R	Top right	Manage project files
Properties Window	F4	Middle right	Set object properties
Form Designer	Double-click form / Shift+F7	Centre	Design the interface
Form Layout	View > Form Layout Window	Bottom right	Set start-up position

Shortcut Keys and Extensions at a Glance

Key / Ext.	Meaning	Key / Ext.	Meaning
F5	Run > Start	Ctrl+T	Components dialog
F4	Properties Window	Ctrl+E	Menu Editor
Ctrl+R	Project Explorer	F7 / F2	Code window / Object Browser
Ctrl+N / Ctrl+O	New / Open Project	.vbp / .vbg	Project / Project Group
.frm / .frx	Form / Form binary data	.bas / .cls / .ctl	Module / Class / User Control

Likely Exam Questions - Fast Answers

Question	Answer
How many menus are in the VB 6.0 Menu Bar?	13
Which toolbar is displayed by default?	Standard
How many icons are in the Toolbox of a Standard EXE?	21 (Pointer + 20 controls)
Which tool is not a control?	Pointer
How to add ActiveX controls to the Toolbox?	Project > Components (Ctrl+T)
Which window shows the tree of forms and modules?	Project Explorer
Extension of a VB project file / form file?	.vbp / .frm
Which key opens the Properties Window?	F4
What does the Properties Window description pane show?	Explanation of the selected property
Which window sets the form start-up position?	Form Layout Window (StartUpPosition = Manual)
Default unit of measurement on the form?	Twip (1440 twips = 1 inch)
Which mode is shown in the Title Bar while designing?	[design]

Commonly Confused - Last-Minute Check

- **Toolbar** = command buttons; **Toolbox** = controls to draw.
- **Form Designer** = design inside the form; **Form Layout** = position of the form on screen.
- **Project Explorer** = files of the project; **Properties Window** = attributes of one object.
- **(Name)** = internal identity; **Caption** = displayed text.
- **.vbp** = project; **.vbg** = group; **.frm** = form; **.frx** = form binary data.

Memory Trick - Order of the IDE

Think top-to-bottom, left-to-right: **Menu** > **Tool**bars > **Tool**box (left) > **Form Designer** (centre) > **Project Explorer**, **Properties**, **Form Layout** (right column, top to bottom).